FastTrack

Phase 1

Team 3

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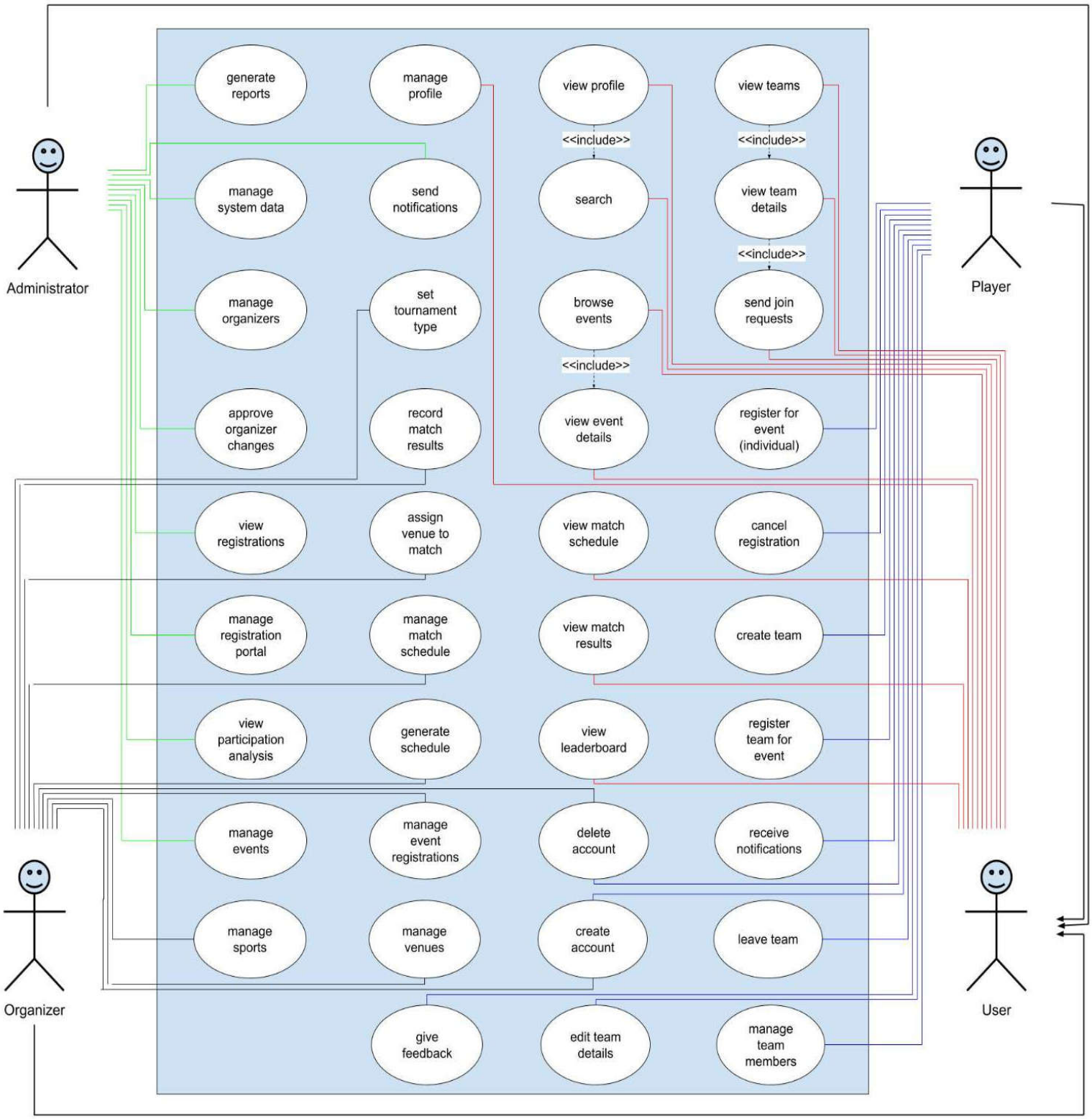
1. Use Case Diagram:



2. Use Cases:

2.1 UC-01

Identifier	UC-01
Name	Create Account
Summary	Allows a new user to register for the FastTrack platform by providing their personal and academic details. Users can register as a Player or an Organizer. Organizer registration requires a special code provided by the admin.
Priority	High
Actor(s)	Organizer, Player
Pre-condition(s)	The user is not currently logged in. The FastTrack application is running.
Post-condition(s)	A new user account is created and stored in the system. The user is logged in and directed to their dashboard.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User selects the "Sign Up" option from the login screen.	
2		Displays the account registration form with account type selection (Player / Organizer).
3	User chooses account type (Player / Organizer).	
4		If Organizer is selected, the system prompts for an admin-provided registration code.
5	User submits the registration form with required details.	
6		Validates the provided information (e.g., email format, password strength, unique roll no.) and, if Organizer, validates the registration code.
7		Confirms that all details are valid and saves the new account to the database.
8		Displays a success message and automatically logs the User into the system, redirecting them to their personalized dashboard.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	User submits the registration form with invalid data (e.g., email already exists, weak password, invalid admin code).	
4		Highlights the specific fields with errors and prompts the User to correct them. The account is not created.

2.2 UC-02

Identifier	UC-02
Name	Delete Account
Summary	Allows a user to permanently remove their account and associated personal data from the FastTrack platform.
Priority	Medium
Actor(s)	Organizer, Player
Pre-condition(s)	The user is logged into the FastTrack application.
Post-condition(s)	The user's account and all associated personal data are permanently removed from the system. The user is logged out and returned to the login screen.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Organizer/Player navigates to their account settings.	
2		Displays the account management options.
3	Organizer/Player selects the "Delete Account" option.	
4		Prompts the Organizer/Player to confirm the deletion, warning that this action is permanent and will remove all their data (e.g., teams, registrations).
5	Organizer/Player confirms their decision to delete the account by re-entering password.	
6		Logs the Organizer/Player out of the system and displays a confirmation message that the account has been successfully deleted.
Alternate Course of Action		

S#	Actor Action	System Response
5	Organizer/Player chooses to cancel the deletion process when prompted for confirmation.	
6		Returns the Organizer/Player to the account settings without any changes.
Alternate Course of Action		
5	Organizer/Player fails to provide correct verification (e.g., wrong password).	
6		Displays an error message indicating incorrect credentials and does not delete the account.

2.3 UC-03

Identifier	UC-03	
Name	Manage Profile	
Summary	Allows a logged-in user to view and update their personal profile information, such as contact details, and password etc.	
Priority	High	
Actor(s)	Administrator, Organizer, Player	
Pre-condition(s)	The user is logged into the FastTrack application.	
Post-condition(s)	The user's profile information is updated in the system.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator/Organizer/Player navigates to their Account Settings page.	
2		Retrieves and displays the Administrator/Organizer/Player's current profile information (e.g., name, email, contact number, profile picture).
3	Administrator/Organizer/Player modifies one or more editable fields (e.g., updates phone number, changes profile picture).	
4	Administrator/Organizer/Player selects the option to Save Changes.	
5		Saves the updated information to the database.

6		Displays a success message confirming the profile has been updated and shows the refreshed information.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Administrator/Organizer/Player attempts to save changes with invalid data (e.g., invalid email format).	
4		Highlights the erroneous fields and prompts the Administrator/Organizer/Player to correct them. The profile is not updated.
Alternate Course of Action		
3	Administrator/Organizer/Player wishes to change their password and selects a "Change Password" option.	
4		System provides a separate form to enter the current password and the new password.
5	Administrator/Organizer/Player enters their current password and a new password, then submits the form.	
6		Validates the input (e.g., current password matches record, new password meets strength criteria, new and confirm fields match).
7		If validation is successful, updates the password in the database. Displays a success message and returns to Account Settings.
8		If validation fails (e.g., wrong current password, weak new password), displays specific error messages and prompts the Administrator/Organizer/Player to correct them. The password is not updated.

2.4 UC-04

Identifier	UC-04
Name	Search
Summary	Allows a user to search for events, teams, or other participants.

Priority	Medium	
Actor(s)	Administrator, Organizer, Player	
Pre-condition(s)	The user is logged into the FastTrack application. The system contains data for events, teams, and users.	
Post-condition(s)	A list of search results matching the user's query and/or filter criteria is displayed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator/Organizer/Player enters a keyword into the global search bar located in the application's header.	
2	Administrator/Organizer/Player presses "Enter" or clicks the "Search" button.	
3		Parses the search query and performs a broad search across all relevant categories (e.g., event names, team names, participant names).
4		Displays a results page, categorizing the findings (e.g., "Events," "Teams," "Participants") that match the search term.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2	Administrator/Organizer/Player clicks on the Filters option next to the search bar.	
3		Displays an expanded search form with multiple filter fields and criteria options.
4	Administrator/Organizer/Player specifies search criteria using available filters.	
5	Administrator/Organizer/Player applies the filters and submits the advanced search.	
6		Constructs a database query incorporating all specified filter criteria.
7		Executes the search and displays filtered results.
8		Displays a message indicating "No results found" if basic or advanced search query yields no matches.

2.5 UC-05

Identifier	UC-05
Name	View Profile
Summary	Allows a user to view profile information for themselves, a team or for another participant they search for.
Priority	High
Actor(s)	Administrator, Organizer, Player
Pre-condition(s)	The user is logged into the FastTrack application. For viewing another's profile, the target user must exist in the system.
Post-condition(s)	The requested profile information is displayed on the screen.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Include UC-04	
2	Administrator/Organizer/Player clicks on a Administrator/Organizer/Player's or team's name in the search results.	
3		Queries the database for the selected Administrator/Organizer/Player's/team's profile data and related information (ensuring privacy controls are respected, e.g., hiding private contact info).
4		Displays a comprehensive profile view for the selected Administrator/Organizer/Player or team.

2.6 UC-06

Identifier	UC-06
Name	Browse Events
Summary	Allows a user to view a list of all available sports events with options to filter them by criteria like sport or status etc.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	The user is logged into the FastTrack application. One or more events have been created in the system.
Post-condition(s)	A list of events with their basic information is displayed to the user.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator/Organizer/Player navigates to the Events of the application.	
2		Retrieves a list of all upcoming and ongoing events from the database.
3	Administrator/Organizer/Player may optionally apply filters (e.g., "Sport: Cricket).	
4		Refreshes the event list to display only those events that match the selected filter or sort criteria.
5	Administrator/Organizer/Player clicks on an event.	
6		Displays the event's key details like name, sport, date, and registration status.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2		If no events are currently available in the system, a message indicating that no events are scheduled at this time is displayed.

2.7 UC-07

Identifier	UC-07
Name	View Event Details
Summary	Allows a user to see detailed information about a specific sports event, such as description, rules, schedule, venue, and registered teams.
Priority	High
Actor(s)	Administrator/Organizer/Player
Pre-condition(s)	The user is logged into the FastTrack application. The specific event exists in the system.
Post-condition(s)	The detailed view of the selected event is displayed.
Typical Course of Action	
S#	**Actor Action**	**System Response**
1	Include UC-06	
2	Administrator/Organizer/Player clicks on event name or View Details button.	

3		Queries the database for all information pertaining to that specific event, including its full description, rules, venue, dates, and a list of registered teams/participants.
4		Displays a comprehensive event details page.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3		If the event ID is invalid or data cannot be retrieved, an error message is displayed.

2.8 UC-08

Identifier	UC-08
Name	View Notifications
Summary	Allows a user to view important updates and alerts from the system, such as registration confirmations, schedule changes, match results, and venue updates.
Priority	Medium
Actor(s)	Administrator/Organizer/Player
Pre-condition(s)	The user is logged into the FastTrack application.
Post-condition(s)	Notifications are presented to the user in their notification inbox.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator/Organizer/Player clicks on the notification bell icon to view their notifications.	
2		Displays a list of recent notifications, with the newest ones highlighted.
3	Administrator/Organizer/Player clicks on a specific notification from the list.	
4		System marks that notification as "read" and unread notifications are displayed above the ones read.
5		If no notifications are found, a friendly message indicating "You're all caught up!" with an empty state illustration is displayed.
Alternate Course of Action		

3	Administrator/Organizer/Player selects "Mark all as read" option.	
4		System updates all unread notifications to "read" status.

2.9 UC-09

Identifier	UC-09	
Name	View match results	
Summary	Allows users to check the final scores and outcomes if complete matches	
Priority	Medium	
Actor(s)	Player, organizer, administrator	
Pre-condition(s)	At least one match must have its result recorded in the system	
Post-condition(s)	Results are displayed to the user	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Selects “Results” from the navigation menu	
2		Displays a list of recently completed sports events and matches
3	Selects a specific match or filters by sport	
4		Displays the final score and winning team/player
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	Filters for a sport with no results	
2		Displays “No results” available for this category

2.10 UC-10

Identifier	UC-10
Name	View leaderboard
Summary	Displays the ranking of players and teams based on points or wins
Priority	Medium
Actor(s)	Guest, player, organizer, administrator
Pre-condition(s)	Ranking data must be calculated and available in

	database	
Post-condition(s)	Current rankings are displayed	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Navigates to the "Leaderboard" section	
2		Displays the overall rankings
3	Applies a filter (e.g., "Cricket", "Badminton")	
4		Updates the table to show rankings for the specific sport

2.11 UC-11

Identifier	UC-11	
Name	Record match results	
Summary	Allows the person managing the sport to unput final scores and winners.	
Priority	High	
Actor(s)	Organizer	
Pre-condition(s)	Organizer is logged in and match status is "Scheduled" or "Ongoing"	
Post-condition(s)	Match status is updated to "Complete" and leaderboard is updated	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Selects a match from the "Manage matches" list	
2		Displays the score entry form
3	Inputs scores for both the sides and selects the winner	
4		Validates the entries
5	Saves the results	
6		Updates match status and notifies relevant players
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Enters non-numeric data in score fields	
4		Displays "Invalid input" and prompts for numbers

2.12 UC-12

Identifier	UC-12	
Name	Set tournament type	
Summary	Defines the logic for match generation (e.g. Knockout, Round Robin)	
Priority	High	
Actor(s)	organizer	
Pre-condition(s)	A sport or event has been created but not yet schedules	
Post-condition(s)	The tournament logic is locked for that specific event	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Selects a sport event to configure	
2		Displays tournament type options
3	Selects “Knockout” or “Round robin”	
4		Saves the configuration to event database

2.13 UC-13

Identifier	UC-13	
Name	Send notifications	
Summary	Broadcasts important updates or alerts to users	
Priority	Medium	
Actor(s)	Organizer, Administrator	
Pre-condition(s)	Actor is logged into an authorized dashboard	
Post-condition(s)	Message is delivered to the target audience	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Accesses the “Notification Center”	
2		Displays the message composer
3	Selects target recipients and types message	
4		Checks for message length and recipients
5	Clicks “Send”	
6		Dispatches the notification via the platform

2.14 UC-14

Identifier	UC-14	
Name	Register for event (individual)	
Summary	Allows a single player to sign up for a solo competition	
Priority	High	
Actor(s)	Player	
Pre-condition(s)	Player is logged in and registration for event is open	
Post-condition(s)	Player is officially listed as a participant	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Selects an individual sport event	
2		Displays "Register" button and event details
3	Clicks "Register"	
4		Checks if player has already registered
5	Confirms registration	
6		Adds player to the list and send confirmation
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Clicks register after the deadline	
4		Displays "Registration Closed"

2.15 UC-15

Identifier	UC-15	
Name	Cancel registration	
Summary	Allows a player to opt out of an event before it begins	
Priority	Low	
Actor(s)	Player	
Pre-condition(s)	Player must be registered for the event and cancellation period is active	
Post-condition(s)	Player is removed from the participant/team list	
Typical Course of Action		
S#	**Actor Action**	**System Response**

1	Navigates to “My registrations”	
2		Display registered events
3	Selects “Cancel registration” for an event	
4		Asks for confirmation
5	Confirms cancellation	
6		Removes the record and frees up the slot

2.16 UC-16

Identifier	UC-16	
Name	Create team	
Summary	Allows a player to form a group for team based sports	
Priority	Medium	
Actor(s)	Player	
Pre-condition(s)	Player is logged in and not currently a captain of another team	
Post-condition(s)	A new team is registered with the player as the leader	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Select “Create team” from the team portal	
2		Display team creation form
3	Enters a team name and selects sport	
4		Validates that the team is unique
5	Submits the form	
6		Creates the team and assigns the player as the captain

2.17 UC-17

Identifier	UC-17
Name	Generate Reports
Summary	Allows an Administrator to generate and export various system reports, such as financial summaries, participation statistics, or tournament results, for analytical purposes.
Priority	Medium

Actor(s)	Administrator	
Pre-condition(s)	The Administrator is logged into the FastTrack application. The system contains relevant data (e.g., match records, registrations) to populate the reports.	
Post-condition(s)	A digital report is generated and displayed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator navigates to the "Reports & Analytics" section of the dashboard.	
		Displays available report categories (e.g., Participation, Financial, Match Results).
2	Administrator selects a specific report type.	
		Displays a form with parameters to customize the report (e.g., date range, specific sport, or tournament name).
3	Administrator specifies the desired filters and clicks "Generate Report."	
		Queries the database to aggregate data based on the selected criteria.
4		Displays the generated report on the screen in a table or graphical format.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Administrator submits filters that yield no matching data.	
		Displays a message indicating "No results found" and prompts the user to adjust criteria.

2.18 UC-18

Identifier	UC-18
Name	Manage System Data

Summary	Allows an Administrator to oversee, backup, and modify core system records, including user logs and database configurations, to ensure system integrity.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	The Administrator is logged into the FastTrack application. The system is operational and the SQL database is accessible.
Post-condition(s)	System data is updated, or a backup is successfully generated and stored.

Typical Course of Action

S#	Actor Action	System Response
1	Administrator navigates to the "System Management" portal.	
		Displays administrative tools: Data Logs, Database Backup, and User Records.
2	Administrator selects the "Database Backup" option.	
		Displays the current database status and the directory for the local backup.
3	Administrator initiates a manual SQL database backup.	
		Performs an automated SQL database backup of all user and event data.
4		Displays a confirmation message once the backup is saved to the local directory.
5	Administrator selects "View Error Logs."	
		Retrieves and displays system errors and critical events with timestamps and severity levels from the local log file.

Alternate Course of Action

S#	Actor Action	System Response
3	Administrator initiates backup but there is insufficient disk space in the local directory.	
		Displays an error message describing the issue and providing an actionable step to resolve it.

2.19 UC-19

Identifier	UC-19
Name	Manage Organizers
Summary	Allows an Administrator to add, view, or remove Organizer accounts and modify their permissions within the platform.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	The Administrator is logged into the FastTrack application. The system contains user records with Organizer roles.
Post-condition(s)	The Organizer's account status or permissions are updated in the database.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator navigates to the "User Management" section and filters for "Organizers."	
		Displays a list of all current Organizers and their active status
2	Administrator selects a specific Organizer to view their details.	
		Queries the database and displays the Organizer's profile, including assigned events and contact info.
3	Administrator selects "Edit Permissions" for the Organizer.	

		Displays a list of toggleable permissions (e.g., Manage Venues, Manage Matches).
4	Administrator updates the permissions and clicks "Save Changes."	
		Saves the updated permissions to the database and displays a success message.
5	Administrator selects an option to "Deactivate Account" for an Organizer who is no longer active	
		Prompts for confirmation and a reason for deactivation.
6	Administrator confirms the deactivation.	
		Updates the account status to "Inactive".
Alternate Course of Action		
S#	**Actor Action**	**System Response**
5	Administrator attempts to deactivate an Organizer who is currently the sole manager of an ongoing event.	
		Displays an error message stating that the Organizer cannot be deactivated until a replacement is assigned.

2.20 UC-20

Identifier	UC-20
Name	Approve Organizer Changes
Summary	Allows an Administrator to review and approve or reject sensitive changes made by Organizers, such as modifications to match schedules or venue assignments, to maintain system oversight
Priority	High
Actor(s)	Administrator

Pre-condition(s)	The Administrator is logged into the FastTrack application. An Organizer has submitted a change request that requires administrative oversight.	
Post-condition(s)	The change request is either applied to the system or discarded. The Organizer is notified of the decision.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator navigates to the "Pending Approvals" dashboard.	
		Displays a list of all pending changes submitted by Organizers. (e.g., Schedule Change, Venue Reassignment).
2	Administrator selects a specific change request to review.	
		Displays the details of the change, showing the original data versus the proposed modification.
3	Administrator reviews the change and selects "Approve."	
		Validates the change (e.g., ensuring a clash-free match schedule) and updates the database.
4		Displays a success message and removes the request from the pending list.
5		Automatically sends a notification to the relevant Organizer regarding the approval.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	The administrator approves a schedule change that creates a conflict.	
		The system detects a schedule clash and displays an error message indicating the specific conflict.

2.21 UC-21

Identifier	UC-21
Name	Manage Registration Portal
Summary	Allows an Administrator to open, close, or modify the settings of the registration portal for various events and tournaments.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	The Administrator is logged into the FastTrack application. One or more events exist in the system.
Post-condition(s)	The registration status (Open/Closed) or deadlines are updated, affecting the Guest user's ability to create accounts or register for events.

Typical Course of Action

S#	Actor Action	System Response
1	Administrator navigates to the "Registration Portal" management interface.	
		Displays a list of all current events and their registration status (e.g., Active, Paused, Closed).
2	Administrator selects a specific event to modify its registration settings.	
		Displays registration parameters including start/end dates, maximum participant capacity, and required fields.
3	Administrator updates the registration deadline or toggles the portal to "Open"	
		Validates that the selected dates are not in the past
4	Administrator clicks "Save Changes."	
		Updates the database and refreshes the portal status for all users.
5		Displays a success message confirming the portal settings have been updated.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Administrator sets a registration deadline that is in the past.	
		Rejects the input and displays an error message stating deadlines cannot be set for past dates.
4	Administrator attempts to open registration for an event that has no assigned venue or schedule.	
		Displays a warning message indicating that event details must be finalized before opening registration.

2.22 UC-22

Identifier	UC-22
Name	View Participation Analytics
Summary	Allows an Administrator to view visual data representations and metrics regarding user engagement, event registration trends, and demographic participation.
Priority	Medium
Actor(s)	Administrator
Pre-condition(s)	The Administrator is logged into the FastTrack application. The system contains participation data from various events and users.
Post-condition(s)	The analytics dashboard is displayed with up-to-date metrics and visualizations.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Administrator navigates to the "Participation Analytics" tab.	
		Queries the database and loads the analytics dashboard
2	Administrator selects a specific metric to view (e.g., "Registrations per Sport").	

		Generates and displays visual charts or graphs representing the data.
3	Administrator applies a date range filter to the data.	
		Refreshes the dashboard visualizations to reflect the new data.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Administrator enters an invalid date range (e.g., start date after end date).	
		Displays a clear error message describing the issue and provides a specific step to resolve it.

2.23 UC-23

Identifier	UC-23
Name	Manage Event
Summary	Allows an Administrator to oversee all aspects of a sports event, including modifying event details, managing participants, and handling global event status..
Priority	High
Actor(s)	Administrator
Pre-condition(s)	The Administrator is logged into the FastTrack application. The specific event exists in the system.
Post-condition(s)	The event details, participant list, or status are updated in the database.
Typical Course of Action	
S#	**Actor Action**	**System Response**
1	Administrator navigates to the "Event Management" dashboard.	
		Displays a list of all current and upcoming events
2	Administrator selects a specific event to manage.	

		Displays comprehensive event details including name, sport, date, and registration status.
3	Administrator modifies event parameters (e.g., changing the venue or updating rules).	
		Validates all date inputs to ensure they are not in the past.
4	Administrator manages the participant list by manually adding or removing teams/players.	
		Updates the event registry and ensures no duplicate team names exist within the same sport
5	Administrator saves the changes.	
		Updates the database and displays a success message confirming the event has been updated
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Administrator attempts to set an event date that has already passed.	
		Rejects the input and displays an error message according to system validation rules.
4	Administrator attempts to add a team name that already exists in that event.	
		Displays an error message indicating the duplicate team name and prevents the addition.

2.24 UC-24

Identifier	UC-24
Name	View Teams

Summary	Allows any user (Player, Organizer, or Administrator) to browse and view a list of all teams registered on the FastTrack platform.	
Priority	Medium	
Actor(s)	Player, Organizer, Administrator	
Pre-condition(s)	The FastTrack application is running and accessible.	
Post-condition(s)	A list of teams is displayed to the user.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to the 'Teams' section from the main menu.	
2		Retrieves and displays a paginated list of all teams with basic info (team name, sport, number of members).
3	User scrolls through or searches the team list.	
4		Updates the displayed list based on any search/filter criteria applied.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2		If no teams found, displays a message indicating that no teams are currently registered on the platform.

2.25 UC-25

Identifier	UC-25	
Name	View Team Details	
Summary	Allows any user (Player, Organizer, or Administrator) to view the detailed information of a specific team, including members, sport, and upcoming events.	
Priority	Medium	
Actor(s)	Player, Organizer, Administrator	
Pre-condition(s)	The FastTrack application is running. At least one team exists in the system.	
Post-condition(s)	The detailed information of the selected team is displayed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Include UC-24	
2	User selects a team from the team list.	

3		Retrieves and displays the full team profile including team name, sport, team lead, members list, and registered events.

2.26 UC-26

Identifier	UC-26	
Name	Register Team for Event	
Summary	Allows a Team Lead (Player) to register their team for a specific sports event on the FastTrack platform.	
Priority	High	
Actor(s)	Player	
Pre-condition(s)	The Player is logged in and is the Team Lead of an existing team. An open event with available slots exists in the system.	
Post-condition(s)	The team is successfully registered for the selected event and confirmation is displayed.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Player navigates to the 'Events' section and selects an open event.	
2		Displays the event details along with a 'Register Team' option.
3	Player selects 'Register Team' and chooses their team from the list.	
4		Validates team eligibility (e.g., sufficient members, event slots available).
5		Registers the team for the event and displays a success message confirming the registration.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
5		If team is not eligible, displays an error message explaining the reason (e.g., 'Insufficient team members', 'Event is full', 'Registration closed') and prevents registration.

2.27 UC-27

Identifier	UC-27
Name	Send Join Request

Summary	Allows a Player to send a request to join an existing team on the FastTrack platform.	
Priority	High	
Actor(s)	Player	
Pre-condition(s)	The Player is logged in and is not currently a member of the target team.	
Post-condition(s)	A join request is sent to the Team Lead and is pending approval.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Include UC-25	
2	Player selects 'Request to Join'.	
3		Checks that the Player is not already a member and the team is not full.
4		Creates a pending join request and notifies the Team Lead.
5		Displays a confirmation message to the Player that the request has been sent.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
4		If the player is already a member or team if full, displays an appropriate message ('Already a member' or 'Team Full') and blocks the request.

2.28 UC-28

Identifier	UC-28	
Name	Leave Team	
Summary	Allows a Player who is a member of a team to voluntarily leave that team on the FastTrack platform.	
Priority	Medium	
Actor(s)	Player	
Pre-condition(s)	The Player is logged in and is a member of at least one team.	
Post-condition(s)	The Player is removed from the team roster and notified of the successful action.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Player navigates to their team details page and selects 'Leave	

	Team'.	
2		Displays a confirmation prompt warning the Player that they will be removed from the team.
3	Player confirms the action.	
4		Removes the Player from the team roster and notifies the Team Lead.
5		Displays a success message confirming the Player has left the team.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
3	Player cancels the confirmation prompt.	
		Returns the Player to the team details page with no changes made.
Alternate Course of Action		
4		If Player is the team lead, displays an error message stating that the Team Lead cannot leave the team without first transferring leadership or disbanding the team.

2.29 UC-29

Identifier	UC-29
Name	Give Feedback
Summary	Allows a Player to submit feedback about an event they participated in on the FastTrack platform.
Priority	Low
Actor(s)	Player
Pre-condition(s)	The Player is logged in and has participated in at least one completed event.
Post-condition(s)	The Player's feedback is saved in the system and visible to event organizers.
Typical Course of Action	
S#	**Actor Action**	**System Response**
1	Player navigates to their event history and selects a completed event.	
2		Displays the event summary along with a 'Give Feedback' option.
3	Player fills in the feedback	

	form (rating and optional comment) and submits.	
4		Validates and saves the feedback, linking it to the Player and the event.
5		Displays a success message confirming the feedback has been submitted.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
4		If feedback already submitted, displays the Player's previously submitted feedback in read-only mode, with no option to submit again.

2.30 UC-30

Identifier	UC-30	
Name	Edit Team Details	
Summary	Allows a Team Lead (Player) to update the details of their team such as team name and description.	
Priority	Medium	
Actor(s)	Player	
Pre-condition(s)	The Player is logged in and is the Team Lead of an existing team.	
Post-condition(s)	The team details are updated in the system and the changes are reflected immediately.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Team Lead navigates to their team details page and selects 'Edit Team'.	
2		Displays the current team details in an editable form.
3	Team Lead modifies the desired fields (e.g., team name, description) and saves changes.	
4		Validates the updated information (e.g., team name uniqueness).
5		Saves the changes and displays a success message.
Alternate Course of Action		

S#	Actor Action	System Response
		If validation fails, highlights the invalid fields with error messages and prevents saving until corrections are made.

2.31 UC-31

Identifier	UC-31	
Name	Manage Team Members	
Summary	Allows a Team Lead (Player) to manage team membership, including approving join requests, removing members, and transferring team leadership.	
Priority	High	
Actor(s)	Player	
Pre-condition(s)	The Player is logged in and is the Team Lead of an existing team.	
Post-condition(s)	The team membership is updated according to the action taken (approved, removed, or leadership transferred).	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Team Lead navigates to the 'Manage Members' section of their team.	
2		Displays a list of current members, pending join requests, and management options.
3	Team Lead selects an action: approve/reject a join request, remove a member, or transfer leadership.	
4		Prompts the Team Lead to confirm the selected action.
5	Team Lead confirms the action.	
6		Executes the action, updates the team roster, and notifies affected members via in-app notification.
7		Displays a success message confirming the action.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
		If team is registered for an active event, displays a warning that the member is part of an active event registration and asks for confirmation before proceeding with removal.

2.32 UC-32

Identifier	UC-8
Name	View Match Schedule
Summary	Allows users to view the schedule of upcoming and ongoing matches for an event.
Priority	High
Actor(s)	Player, Organizer, Administrator
Pre-condition(s)	System is running and matches are scheduled.
Post-condition(s)	Match schedule displayed.

Typical Course of Action

S#	Actor Action	System Response
1	User opens Match Schedule	
2		System retrieves matches
3	User optionally filters event	
4		System displays date, time and venue

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		System shows 'No matches scheduled' if empty

2.33 UC-33

Identifier	UC-33
Name	View Registrations
Summary	Allows organizers or administrators to view participants registered for events.
Priority	High
Actor(s)	Organizer, Administrator
Pre-condition(s)	User logged in.
Post-condition(s)	Registration list displayed.

Typical Course of Action

S#	Actor Action	System Response
1	User opens Registrations	
2		System lists events
3	User selects event	
4		System displays registered players/teams

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		System displays 'No registrations found'

2.34 UC-34

Identifier	UC-34
Name	Manage Sports
Summary	Administrator can add, update, or delete sports.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	Admin logged in.
Post-condition(s)	Sports database updated.

Typical Course of Action

S#	Actor Action	System Response
1	Admin opens Manage Sports	
2		System lists sports
3	Admin edits/adds sport	
4		System validates and saves

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		Validation error if data invalid

2.35 UC-35

Identifier	UC-35
Name	Manage Venues

Summary	Administrator manages venues used for matches.
Priority	High
Actor(s)	Administrator
Pre-condition(s)	Admin logged in.
Post-condition(s)	Venue information updated.

Typical Course of Action

S#	Actor Action	System Response
1	Admin opens Manage Venues	
2		System lists venues
3	Admin adds/edits venue	
4		System saves venue

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		Error if duplicate venue

2.36 UC-36

Identifier	UC-25
Name	Manage Event Registrations
Summary	Organizer approves or rejects event registrations.
Priority	High
Actor(s)	Organizer, Administrator
Pre-condition(s)	Registration requests exist.
Post-condition(s)	Registration status updated.

Typical Course of Action

S#	Actor Action	System Response
1	User opens Event Registrations	
2		System lists requests
3	User selects request	
4		System shows details
5	User approves/rejects	
6		System updates status

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		System shows 'No pending requests'

2.37 UC-37

Identifier	UC-37
Name	Generate Schedule
Summary	Organizer generates tournament schedule automatically.
Priority	High
Actor(s)	Organizer
Pre-condition(s)	Participants exist.
Post-condition(s)	Schedule generated.

Typical Course of Action

S#	Actor Action	System Response

1	Organizer opens generator	
2		System lists events
3	Organizer selects event	
4		System generates schedule

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		Error if insufficient participants

2.38 UC-38

Identifier	UC-38
Name	Manage Match Schedule
Summary	Organizer edits match timing or date.
Priority	High
Actor(s)	Organizer
Pre-condition(s)	Schedule exists.
Post-condition(s)	Schedule updated.

Typical Course of Action

S#	Actor Action	System Response
1	Organizer opens schedule manager	
2		System lists matches
3	Organizer selects match	
4		System allows editing
5	Organizer updates time/date	
6		System saves changes

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		Conflict detected — update rejected

2.39 UC-39

Identifier	UC-28

Name	Assign Venue to Match
Summary	Organizer assigns venue to scheduled matches.
Priority	High
Actor(s)	Organizer
Pre-condition(s)	Matches and venues exist.
Post-condition(s)	Venue assigned.

Typical Course of Action

S#	Actor Action	System Response
1	Organizer opens match schedule	
2		System lists matches
3	Organizer selects match	
4		System shows venues
5	Organizer selects venue	
6		System saves assignment

Alternate Course of Action

S#	Actor Action	System Response
Alt-1		Error if venue unavailable

3. Analysis Class Diagram:

4. Non-Functional Requirements:

1. FastTrack shall load the dashboard within 2 seconds on a local machine for a logged-in user.

2. FastTrack shall generate a clash-free match schedule for a tournament with 20 teams in under 10 seconds.

3. FastTrack shall load the match schedule screen within 2 seconds for a tournament containing 50 matches.

4. FastTrack shall return search results for a basic query in under 1 second.

5. FastTrack shall refresh the participation analytics dashboard within 5 seconds after data updates.

6. FastTrack shall export tournament data to a CSV file containing up to 1,000 records within 5 seconds.

7. The SQL database shall query and display the leaderboard from a table of 200,000+ match records in less than 3 seconds.

8. FastTrack's CSV export feature shall successfully generate a file for a tournament containing 500+ matches.

9. A new user (Player) shall be able to create an account and register for an individual event within 3 minutes.

10. FastTrack shall use consistent UI elements for similar actions across all modules.

11. All error messages shall clearly describe the issue and provide a specific, actionable step to resolve it.

12. FastTrack shall be fully usable on a standard tablet screen (resolution 768×1024), with no horizontal scrolling required for primary tasks.

13. FastTrack shall be fully functional on the latest two major versions of Google Chrome, Mozilla Firefox and Apple Safari.

14. FastTrack shall store all passwords using a strong hashing algorithm with a salt.

15. FastTrack shall prevent a user with the "Player" role from accessing any interface elements designated for "Organizer" or "Administrator" functions.

16. FastTrack shall reject all form inputs containing malicious scripts (e.g., <script> tags) with a generic error message.

17. FastTrack shall automatically log out a user after 15 minutes of inactivity.

18. FastTrack shall ensure a player's contact information is not visible to other players unless explicitly required for team management.
19. FastTrack shall maintain 99.5% functionality during the week of the sports festival with no critical crashes.

20. FastTrack shall perform an automated SQL database backup of all user and event data

every 24 hours to a local directory.

21. In the event of a crash, FastTrack shall restart and be fully operational within 5 minutes with no data loss.

22. FastTrack shall save a user's progress (e.g., forming a team, filling a registration form) as a draft every 1 minute.

23. FastTrack shall log all system errors and critical events with a timestamp and severity level to a local log file.

24. The source code for the clash-free scheduler module shall have a minimum of 85% test coverage.

25. All code committed to the repository shall adhere to the project's defined ESLint/code style rules without any errors.

26. The scheduling algorithm shall be a distinct, replaceable module with a well-defined interface.

27. The system's architecture shall support scaling the database to 5,000 registered users without requiring a redesign.

28. FastTrack shall validate all date inputs (e.g., match schedules, registration deadlines) to ensure they are not in the past before accepting them.

29. FastTrack shall prevent duplicate team names within the same sport event.

30. FastTrack shall enforce strong password requirements for all user accounts, requiring a minimum of 8 characters, including at least one uppercase letter, one lowercase letter, one number, and one special character.

31. FastTrack shall ensure all usernames are unique across the system and reject any registration attempt with a duplicate username, prompting the user to choose a different one.